SMART PMB

DIGITAL PADDY ECOSYSTEM FOR SRI LANKA

PROJECT PROPOSAL

A Ministry-Sponsored Demonstration and Controlled Pilot

Submitted for consideration by
The Ministry responsible for agriculture and the Paddy Marketing Board (PMB), Sri Lanka

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Version 1.0 • Discussion draft

This proposal builds on the Smart PMB concept paper dated 14 May 2026 and the current web demonstrator. It is a basis for discussion—not a procurement specification, production certification or cost commitment.

Executive summary

Smart PMB is a proposed digital platform to connect farmers, authorized purchasers, rice mill operators, warehouse managers, drivers, PMB officers and system administrators around trusted paddy-purchasing, storage and transport records. It is intended to reduce paper-based duplication, improve service visibility and give decision-makers a timelier view of operations—from a farmer's submission through verification, warehousing and delivery. The platform also provides a basis for traceable lots, role-specific dashboards and carefully governed public reporting of aggregate information.

A web-based demonstrator has been developed using the maintained Smart PMB application. It includes role-based portals and representative workflows for farmer services, PMB officer operations, warehouse management, transport, partner licensing, reporting and public traceability. The demonstrator is not represented as a production-certified national service. This proposal therefore seeks Ministry and PMB sponsorship for a discovery phase and a controlled pilot, with security, data governance, accessibility, integration and operating readiness validated before wider deployment.

The recommended approach is incremental: confirm requirements and hosting; prepare a non-production pilot environment; validate selected workflows with representative users; measure outcomes against an agreed baseline; and scale only after formal acceptance. Advanced AI capabilities such as yield prediction, crop-disease recognition and route optimization are potential later workstreams—not current production claims—and should proceed only when suitable data, independent validation and human oversight are in place.

Proposal at a glance

Item	Proposal
Purpose	Validate a secure, integrated PMB digital workflow through discovery and a controlled pilot.
Users	Farmers, purchasers, mill operators, warehouse managers, drivers, PMB officers and administrators.
Delivery model	Progressive phases with Ministry/PMB ownership, user acceptance and explicit go/no-go gates.
Indicative duration	Approximately 5–8 months from mobilization to a pilot evaluation; subject to procurement, data access and hosting decisions.
Funding request	Approve discovery and pilot planning; prepare a costed implementation plan after requirements and hosting are confirmed.

Decision sought

Nominate a Ministry sponsor and PMB product owner, authorize a time-boxed discovery phase, and identify candidate pilot sites and user groups.

1. Case for change

Paddy purchasing, storage and distribution involve multiple actors and locations. Where records and communications are fragmented, it is harder to verify transactions, reconcile stock, follow delivery progress and prepare consistent reports. Smart PMB proposes one governed digital workflow while retaining PMB authority over policy, pricing and operational decisions.

Current challenge	Opportunity for PMB
Fragmented records	Manual forms, spreadsheets and disconnected processes make it difficult to maintain one consistent view of purchases, stock, licences and deliveries.
Limited transaction visibility	Farmers and officers may have difficulty following the status and evidence for a harvest or purchase from submission to completion.
Delayed operational information	Warehouse stock, capacity and transport information can be slow to consolidate across locations.
Uneven access to information	Farmers may not have timely access to current guaranteed prices, status updates or a reliable digital record of a transaction.
Reporting effort	National and regional summaries can require repeated manual compilation, increasing delay and the risk of inconsistent figures.

Project objectives

- Establish a secure, role-based digital record for agreed PMB workflows.
- Improve the traceability and status visibility of harvest, purchase, warehouse and delivery records.
- Support more timely operational decisions through consistent dashboards and reports.
- Make services easier to reach through responsive web interfaces and accessible notifications.
- Improve accountability through permission controls, recorded decisions and audit trails.
- Create a governed foundation for future data analytics without automating public decisions prematurely.

Expected public value

If the pilot validates the workflows and service model, Smart PMB can support more timely farmer information, better transaction traceability, improved stock and logistics visibility, reduced duplicate data entry and more reliable operational reporting. Benefits must be measured against a pre-pilot baseline; no savings or service improvement is assumed without evidence.

2. Proposed solution and scope

Smart PMB is a role-based digital service with a central operational record and distinct views for each authorized user group. A typical workflow moves from registration and record creation to officer review, stock handling, delivery and reporting.

Step	How the workflow works
1. Register and access	A user signs in through an account and role appropriate to their responsibilities.
2. Submit or record	A farmer or authorized operator creates a harvest, purchase, request, licence or delivery record.
3. Review and decide	An authorized PMB officer or manager reviews the record, records a decision and—where required—adds a reason.
4. Store and move	Approved stock is associated with a warehouse; transport staff manage assigned routes and delivery status.
5. Verify and report	A traceable lot reference, operational dashboards and approved aggregate reports support follow-up and oversight.

Demonstrator, pilot and later phases

Stage	Scope and status
Available in the current demonstrator	Role-specific web portals; farmer and harvest workflows; PMB officer approvals and pricing; warehouses and stock records; vehicles, routes and deliveries; partner/mill licensing workflows; reports; QR lot traceability; public aggregate transparency; selected English/Sinhala interface coverage.
To validate and complete for a pilot	Ministry-approved operating procedures; user and data onboarding; pilot-site configuration; complete Sinhala/Tamil localization plan; data-quality checks; security and privacy review; user training; backup/restore and monitoring; integration decisions; user acceptance testing and support arrangements.
Later phase / separate approval	Native mobile application; validated yield or disease models; advanced route optimization; satellite/IoT systems; banking or government payment integrations; nationwide data migration and scale-out; blockchain or export/tax integrations.

Scope boundary

The pilot covers digital record-keeping and payment-status visibility, not the construction of banking infrastructure or direct farmer fund disbursement. National rollout, legacy data migration and major third-party integrations require separate approval.

3. Functional requirements

The following requirements define the service outcomes to confirm during discovery. “Must” items are proposed pilot essentials; “Should” items are desirable subject to readiness; “Later” items require a separate data, governance and funding decision.

ID / priority	Requirement and proposed acceptance
FR-01 • Must	Identity, roles and permissions The system shall authenticate users and restrict functions and records according to approved roles, including Farmer, Authorized Purchaser, Mill Owner, PMB Officer, Warehouse Manager, Driver and Administrator. Account creation, approval and access changes shall be controlled and reviewable.
FR-02 • Must	Farmer profile and service access A farmer shall be able to maintain an approved profile, submit harvest information, view applicable guaranteed prices, follow harvest and payment-record status, and receive relevant service notifications. Access to another farmer’s information shall be denied unless explicitly authorized.
FR-03 • Must	Purchasing and harvest workflow Authorized users shall record harvest/purchase information using agreed fields for date, paddy type, quantity, quality/grade, location and price. Officers shall review and record the permitted outcome (for example, verified, collected or rejected) with a reason where required.
FR-04 • Must	Price and transaction evidence Authorized officers shall maintain the applicable paddy types and guaranteed-price records. The system shall retain a dated transaction record and expose the relevant status to permitted users. The pilot covers payment-status records; it does not assume direct bank disbursement.
FR-05 • Must	Digital receipt and traceability For an eligible collected lot, the system shall create a unique reference and QR code that opens a public, privacy-safe trace page. The trace page shall show only approved lot-journey information and shall not expose personal farmer or financial records.
FR-06 • Must	Warehouse and inventory control Authorized warehouse users shall view stock by agreed categories, see capacity indicators, record approved stock movements and review transaction history. Adjustments shall include actor, timestamp, reason and any required approval; alerts shall be configurable.

3. Functional requirements (continued)

ID / priority	Requirement and proposed acceptance
FR-07 • Should	Partner and mill administration Applicants shall be able to submit required partner or mill details and supporting information. Authorized officers shall review applications, record inspection outcomes, update status and maintain relevant milling or allocation records.
FR-08 • Must	Transport and delivery Authorized officers shall manage vehicles, reusable routes and deliveries, assign drivers and destinations, and review delivery status. Drivers shall see assigned tasks and record permitted responses and location updates. Location tracking shall be purpose-limited and clearly governed.
FR-09 • Must	Dashboards and reports Users shall see role-appropriate summaries. Authorized staff shall filter and export agreed operational reports. A public dashboard shall publish aggregate figures only after the PMB approves definitions, refresh cadence and disclosure controls.
FR-10 • Should	Communication and languages The system shall provide in-app notices and support approved email/SMS notifications where configured. The pilot shall validate English and Sinhala coverage and define a reviewed Tamil localization plan for public and operational journeys.
FR-11 • Must	Administration and oversight Authorized administrators shall manage users, roles and permissions; review audit and login activity; configure approved operational controls; and support service continuity. Administrative preview functions shall use synthetic data and remain distinct from real-user impersonation.
FR-12 • Later	Analytics and decision support The demonstrator includes trend and capacity indicators. Any advanced yield, price, disease or logistics model shall be a separately approved enhancement, tested against representative Sri Lankan data and presented as decision support—not an automated basis for eligibility, price-setting or enforcement.

Important distinction

The current demonstrator contains trend and warehouse-capacity indicators. It does not claim validated AI for yield, disease detection or route optimization. Those capabilities require a separate, evidence-led workstream and human decision-making.

4. Non-functional requirements

These are proposed acceptance criteria for planning. Numerical targets below are initial discussion targets—not measured performance or an existing service-level commitment—and must be validated with PMB ICT, security and operations.

ID	Quality requirement and proposed acceptance
NFR-01	Security Use least-privilege role access, secure password hashing, protected session/token handling, HTTPS, managed secrets, dependency patching and security testing. Staff access controls and privileged actions must be reviewed before pilot launch.
NFR-02	Privacy and data governance The Ministry/PMB shall define data ownership, purposes, permitted access, retention, correction, disclosure and deletion rules. Minimize collection of personal data; protect identity, payment-status and location data; complete legal/privacy review before live data is loaded.
NFR-03	Performance Proposed pilot target: routine pages and API operations complete within 3 seconds at the 95th percentile under an agreed pilot workload and network profile. Confirm through load testing; this is a target, not a measured current SLA.
NFR-04	Availability and recovery Proposed initial target: 99.5% monthly service availability during pilot operations, excluding approved maintenance. Daily backups and tested restore procedures; discuss initial RPO ≤24 hours and RTO ≤8 hours with PMB ICT before contracting.
NFR-05	Usability and accessibility Responsive web journeys shall work on common desktop and smartphone browsers. Conduct task-based testing with farmers and officers, including lower-bandwidth conditions. Target WCAG 2.1 AA practices; confirm through an accessibility review.
NFR-06	Localization Support approved Sinhala, Tamil and English terminology for priority user journeys. Text, date, number and currency formats must be consistent; translations should be reviewed by competent speakers and operational stakeholders.
NFR-07	Scalability and interoperability The design shall support progressive multi-district growth and documented APIs. Data exports shall use agreed, usable formats. Hosting, integration boundaries and data residency must be decided by the Ministry/PMB before production.
NFR-08	Reliability and data integrity Validate required fields and permitted transitions; protect multi-step writes from partial completion; prevent duplicate submissions where feasible; maintain traceable, consistent records and routine monitoring.
NFR-09	Auditability Record who performed key approvals, stock changes, price updates, account/permission changes and administrative actions, with timestamp and relevant before/after values or reason. Apply approved retention and tamper-protection controls.
NFR-10	Maintainability and support Use documented configuration, version control, automated checks, staged release and rollback procedures. Define operational ownership, incident response, support hours and change control before a live pilot.

Security and data-governance principles

- The Ministry/PMB remains the owner and decision-maker for operational data, access policy, retention and permitted disclosure.
- Use least privilege; limit collection of personal information; protect identifiers, contact details, payment statuses and location data.
- Confirm applicable Sri Lankan privacy, public-sector security, procurement and records-management requirements through Ministry legal and ICT review.
- Complete threat modelling, vulnerability review, restore testing, incident response planning and user training before live pilot data is introduced.

5. Technology approach and why it is suitable

The recommendation is to extend the current working demonstrator rather than replace it with a second, parallel stack. The technologies below describe the maintained application; final hosting and integrations remain subject to Ministry approval.

Layer	Current technology	Why it fits the pilot
Web application	Next.js / React / TypeScript	Responsive role-based web portals and a maintainable user interface; one browser-based service supports desktop and smartphone access without requiring a native mobile app for the first pilot.
Business API	Python / Django / Django REST Framework	Centralizes domain rules, validation, role checks and reporting endpoints. It is the current backend and can be extended in a controlled way as PMB workflows are confirmed.
Operational data	PostgreSQL	Relational data storage supports linked records, consistency constraints and reporting across farmers, transactions, warehouses, routes and deliveries. Final hosting must meet Ministry security and data-residency requirements.
Authentication and access	JWT-based sessions, protected cookies, application RBAC	Allows user-specific access across portals; permissions are checked in the web layer and API. The Ministry should approve session lifetime, MFA requirements and account lifecycle policy.
Maps and tracking	Google Maps integration (optional, subject to key and policy)	Supports map-based vehicle views and route-related user journeys. Any production use requires API-key management, privacy review, provider terms and cost approval.
Traceability and reporting	QR generation; PDF and dashboard reporting	Provides scannable lot references and readable operational summaries without requiring users to inspect raw database records.
Notifications	In-app notices; optional email/SMS adapters	Keeps users informed through selected channels. SMS provider, delivery fees and message content require PMB approval and privacy review.

Technology rationale and decisions

- The proposed pilot should build on the demonstrator’s existing Next.js + Django + PostgreSQL architecture rather than introduce a second API stack without a demonstrated need.
- The core frameworks are widely used and open source, reducing framework licence dependence while still requiring budget for hosting, security, communications, support and training.
- A browser-based first release lowers installation and device-management effort; a native Flutter application can be assessed later if pilot evidence shows a clear need for offline or device-specific capabilities.
- The original concept paper lists Flutter, Node.js/Express, Firebase, AWS and Azure. Those were conceptual options; the current maintained demonstrator uses Next.js/React, Django/DRF and PostgreSQL. Final hosting remains a Ministry decision.
- The repository includes Stripe test-mode functionality for an internal system-change workflow. This proposal does not recommend Stripe for farmer payments or public funds; any payment integration requires a separate Ministry-led assessment.

Hosting decision

The demonstrator uses PostgreSQL configuration and the project documentation references a managed PostgreSQL service. For a Ministry pilot, hosting, data residency, backup location, administrator access and supplier terms must be reviewed and approved before production data is used.

6. Delivery plan and phase gates

A staged Agile delivery is recommended so that operational users can review working flows early. Each phase ends with a documented decision to proceed, remediate, extend or stop. Indicative duration is approximately 20–32 weeks, excluding procurement and any major external integration.

Phase	Duration	Key outputs / exit evidence
0. Mobilize and discover	2–4 weeks	Name Ministry/PMB owners; confirm workflows, pilot goals, stakeholder groups, data classifications, constraints and evaluation baseline.
1. Pilot design and readiness	4–6 weeks	Confirm requirements and hosting; review privacy/security; map data; prepare test environment; agree integrations, support model and acceptance criteria.
2. Configure and validate	4–6 weeks	Adapt priority workflows; load approved test data; conduct functional, integration, accessibility and security checks; complete user acceptance testing.
3. Controlled field pilot	8–12 weeks	Train pilot users; operate in selected PMB locations; provide support; monitor service, adoption, data quality and agreed operational KPIs.
4. Evaluate and decide scale	2–4 weeks	Compare results with baseline; document issues, costs and user feedback; approve remediation and a costed, phased rollout plan or stop/extend pilot.

Minimum phase gates

- Discovery sign-off: approved workflows, roles, data definitions, pilot sites, service boundaries and baseline measures.
- Readiness sign-off: approved hosting and privacy/security plan; test environment; support and training plan; data is synthetic or explicitly approved.
- UAT sign-off: representative users complete agreed scenarios with no unresolved critical defects and accessibility/security issues assigned.
- Pilot exit: evidence reviewed against KPIs, risks, user feedback and operating cost before any scale-up decision.

7. Governance, roles and risk management

Role	Responsibility
Ministry sponsor / steering group	Set public outcomes, resolve cross-agency decisions, approve phase gates and hosting/security direction.
PMB product owner	Own workflows, priorities, data definitions, acceptance criteria and operational readiness.
PMB district and site focal points	Validate field processes, nominate users, coordinate training and report pilot issues.
PMB ICT / security / privacy leads	Review architecture, access controls, data handling, hosting, continuity, integration and incident response.
Delivery team	Configure, test, document and support the pilot; report issues transparently and maintain a change log.
Representative users	Farmers, purchasers, mill operators, warehouse managers, drivers and officers participate in co-design and UAT.

Key risks and mitigations

Risk	Proposed mitigation
Data quality and legacy records	Profile and cleanse agreed pilot data; define identifiers and ownership; migrate only approved, necessary records; reconcile totals with PMB.
Adoption and digital access	Co-design with field users; provide Sinhala/Tamil support plan, demonstrations and assisted channels; retain an approved fallback while pilot issues are addressed.
Connectivity constraints	Test real pilot-site networks; optimize common screens; document offline/manual contingency. Offline synchronization is not claimed for the current demonstrator.
Privacy or cyber incident	Complete threat/privacy reviews, least-privilege access, backups, monitoring, incident exercises and independent security testing before production use.
Integration / hosting uncertainty	Decide hosting and integration ownership early; use a sandbox first; budget for cloud, SMS, map, email and support dependencies.
Premature AI use	Keep predictions advisory; document data limitations, validation and human review; do not automate price, eligibility or enforcement decisions in the pilot.

A steering group should meet at least monthly during discovery and the pilot. Operational priorities and issues should be reviewed weekly by the PMB product owner and delivery lead. Scope changes, data access and production releases should be logged and approved through the agreed governance process.

8. Outcomes, measurement and investment

The pilot should establish a baseline before go-live and compare results at agreed intervals. PMB should set target values only after baseline data and operational constraints have been reviewed.

Measure	How it will be assessed
Digital transaction coverage	Share of eligible pilot purchases/harvests captured end-to-end in the system.
Processing time	Median time from farmer submission to officer decision and from decision to recorded collection.
Stock accuracy	Difference between system stock and agreed physical reconciliation by warehouse and stock category.
Delivery performance	Share of pilot deliveries completed within the agreed schedule; reasons for delay/rejection.
Traceability coverage	Share of eligible collected lots with a valid QR/lot record and successful trace lookup.
User experience and access	Task completion, support requests, user satisfaction and notification delivery across user groups/languages.
Service reliability	Availability, response-time distribution, incident count and recovery performance against agreed pilot targets.

Budget categories to cost after discovery

Cost area	Items to estimate
Implementation and change	Workflow discovery, configuration, integration, testing, project management and documentation.
Hosting and data services	Application hosting, managed database, backups, monitoring, domain/TLS and recovery environment.
Communications and maps	SMS/email usage, map API consumption and any approved third-party service fees.
Security and assurance	Independent security testing, privacy review, accessibility review and remediation.
People and operations	Training, field support, help desk, data stewardship, maintenance and ongoing service ownership.

The use of open-source frameworks can reduce software licence dependence but does not make the service cost-free. A detailed budget should include hosting and data services, communications and map usage, security assurance, integration, migration, training, field support and ongoing maintenance. No unsupported currency estimate is included in this proposal.

9. Ministry demonstration plan

A short, guided demonstration (approximately 8–10 minutes) can show how the roles and workflow connect. Use a prepared staging environment and synthetic records; do not use real farmer, payment or GPS data in a public demonstration.

Demo segment	What the Ministry can observe
Officer view	Sign in as a PMB Officer and review role-appropriate dashboard and work queues.
Route and delivery	Create a sample route (origin, destination, distance, optional warehouse), schedule a delivery and show assignment/status workflow.
Farmer transaction	Submit or review a sample harvest, demonstrate an approval decision and show the resulting lot/receipt record.
Warehouse control	Review sample stock/capacity, record a controlled stock movement and show its audit entry.
Public assurance	Scan a synthetic QR lot and open the aggregate transparency page; demonstrate that personal data is not exposed.
Governance	Show role permissions and audit history using non-production data.

Decision requested from the Ministry and PMB

- Nominate a Ministry sponsor and PMB product owner with authority to confirm operational workflows.
- Authorize a time-boxed discovery and controlled pilot planning phase, subject to procurement and budget rules.
- Select candidate pilot locations and representative users; approve a process for sanitized test data and any subsequent live data.
- Direct PMB ICT, security and privacy leads to review hosting, data ownership, retention, access, integration and continuity requirements.
- Agree pilot KPIs and phase-gate acceptance criteria before field use; request a detailed costed rollout plan after discovery.

Recommended next step

Approve a short discovery and pilot-design phase. Return to the Ministry with a validated requirements baseline, security/hosting proposal, selected pilot sites, measurement plan and costed implementation schedule before authorizing live data or a wider rollout.

Assumptions and limitations

- This proposal describes a demonstrator and a path to a controlled pilot; it is not a claim that the system is certified, fully deployed nationwide, or already integrated with PMB legacy systems.
- Current demonstrator capabilities and proposed future capabilities are separated in this document. Advanced AI, a native mobile app, offline synchronization, banking integration and national rollout require separate design and approval.
- Service-level targets, hosting, data retention, localization coverage and budget figures are proposals for discussion and must be validated with Ministry/PMB stakeholders.
- The pilot should use synthetic or approved data until the Ministry authorizes live personal and operational data processing.

Source basis

This proposal is based on the user-supplied “SMART PMB — Digital Paddy Ecosystem for Sri Lanka” project plan (Group D, LNBTI, 14 May 2026) and the maintained Smart PMB repository and demonstrator reviewed on 27 October 2026. Requirements, hosting decisions, legal obligations, cost and service targets remain subject to Ministry/PMB validation.